

Derivation of the Volume Formula for Cylinders

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A **cylinder** is a three-dimensional figure with two congruent and parallel **circular bases**.

Base Area

The **base area** of a cylinder is the area of a circle, given by

$$A = \pi r^2$$

where r is the radius of the circular base.

Height

The **height** is the perpendicular distance between the two circular bases.

$$\text{Height} = h$$

Volume of a Cylinder

The formula for the volume of a cylinder is

$$V = \pi r^2 h$$

The volume is the area of the circular base multiplied by the height.

Cubic Units

The units of volume are expressed in **cubic units**.

Examples include

cm^3 , m^3 , in^3

Examples

Example 1

A cylinder has a radius of 4 cm and a height of 10 cm. Find its volume in terms of π .

$$V = \pi r^2 h$$

$$V = \pi(4)^2(10) = 160\pi$$

Therefore, $V = 160\pi \text{ cm}^3$.

Example 2

A cylindrical container has a radius of 6 cm and a height of 8 cm. Determine its volume.

$$V = \pi(6)^2(8)$$

$$V = 288\pi$$

Therefore, $V = 288\pi \text{ cm}^3$.

Example 3

A can has a diameter of 12 cm and a height of 15 cm. Find its volume.

$$r = \frac{12}{2} = 6 \text{ cm}$$

$$V = \pi(6)^2(15) = 540\pi$$

Therefore, $V = 540\pi \text{ cm}^3$.

Practice Exercises

Practice 1

A cylinder has a radius of 4 cm and a height of 10 cm. Find its volume in terms of π .

Practice 2

A cylindrical container has a radius of 6 cm and a height of 8 cm. Determine its volume.

Practice 3

Find the volume of a cylinder with radius 9 m and height 5 m.

Practice 4

A can has a diameter of 12 cm and a height of 15 cm. Find its volume.

Practice 5

A water tank has a radius of 3.5 m and a height of 12 m. Determine its volume. Express your answer in terms of π .

Practice 6

A cylinder has a volume of $392\pi \text{ cm}^3$ and a radius of 7 cm. Find its height.

Practice 7

A cylinder has a volume of $450\pi \text{ m}^3$ and a height of 18 m. Find its radius.

Practice 8

The radius of a cylinder is doubled while its height remains unchanged. How many times larger is the new volume than the original?

Activities

Activity 1

A cylinder has a radius of 5 cm and a height of 9 cm. Find its volume in terms of π .

Activity 2

A cylindrical container has a radius of 7 cm and a height of 6 cm. Determine its volume.

Activity 3

Find the volume of a cylinder with radius 8 m and height 11 m.

Activity 4

A can has a diameter of 16 cm and a height of 9 cm. Find its volume.

Activity 5

A storage tank has a radius of 4 m and a height of 10 m. Determine its volume. Express your answer in terms of π .

Activity 6

A cylinder has a volume of $616\pi \text{ cm}^3$ and a radius of 7 cm. Find its height.

Activity 7

A cylinder has a volume of $784\pi \text{ m}^3$ and a height of 16 m. Find its radius.

Activity 8

The radius of a cylinder is tripled while its height remains unchanged. How many times larger is the new volume than the original?

Thank you!